Chpater1 Mechanics & Fluid Dynamics

Lesson6 Fluid Dynamics, Continuity & Viscosity

1. The spring of the pressure gauge shown in **Figure P9.20** has a force constant of $1\ 250\ \mathrm{N/m}$, and the piston has a radius of $1.20\ \mathrm{cm}$. As the gauge is lowered into water, what change in depth causes the piston to move in by $0.750\ \mathrm{cm}$?

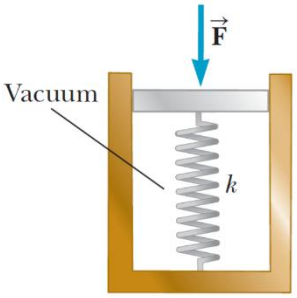


Figure P9.20

2. **(a)** Calculate the absolute pressure at the bottom of a fresh-water lake at a depth of $27.5\ \mathrm{m}$. Assume the density of the water is $1.00 \times 103\ \mathrm{kg/m^3}$ and the air above is at a pressure of $101.3\ \mathrm{kPa}$. **(b)** What force is exerted by the water on the window of an underwater vehicle at this depth if the window is circular and has a diameter of $35.0\ \mathrm{cm}$?

3. Mercury is poured into a U-tube as shown in **Figure P9.22a** on page 324. The left arm of the tube has cross-sectional area A_1 of $10.0\ \mathrm{cm^2}$, and the right arm has a cross-sectional area A_2 of $5.00\ \mathrm{cm^2}$. One hundred grams of water are then poured into the right arm as shown in **Figure P9.22b**. **(a)** Determine the length of the water column in the right arm of the U-tube. **(b)** Given that the density of mercury is $13.6\ \mathrm{g/cm^3}$, what distance h does the mercury rise in the left arm?

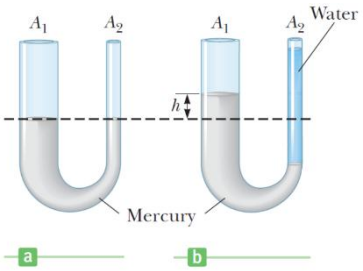


Figure P9.22

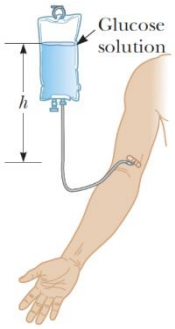


Figure P9.23

4. A collapsible plastic bag **(Figure P9.23)** contains a glucose solution. If the average gauge pressure in the vein is $1.33 \times 10^3\ \mathrm{Pa}$, what must be the minimum height h of the bag in order to infuse glucose into the vein? Assume the specific gravity of the solution is 1.02.

5. The deepest point in the ocean is in the Mariana Trench, about $11\ \mathrm{km}$ deep. The pressure at the ocean floor is huge, about $1.13 \times 10^8\ \mathrm{N/m^2}$. **(a)** Calculate the change in volume of $1.00\ \mathrm{m^3}$ of water carried from the surface to the bottom of the Pacific. **(b)** The density of water at the surface is $1.03 \times 10^3\ \mathrm{kg/m^3}$. Find its density at the bottom.

6. A container is filled to a depth of $20.0\ \text{cm}$ with water. On top of the water floats a 30.0-cm-thick layer of oil with specific gravity 0.700. What is the absolute pressure at the bottom of the container?

7. A table-tennis ball has a diameter of $3.80\ \text{cm}$ and average density of $0.084\ 0\ \text{g/cm}^3$. What force is required to hold it completely submerged under water?

8. A small ferryboat is $4.00\ \text{m}$ wide and $6.00\ \text{m}$ long. When a loaded truck pulls onto it, the boat sinks an additional $4.00\ \text{cm}$ into the river. What is the weight of the truck?

9. A wooden block of volume $5.24 \times 10^{-4}\ \text{m}^3$ floats in water, and a small steel object of mass m is placed on top of the block. When $m = 0.310\ \text{kg}$, the system is in equilibrium, and the top of the wooden block is at the level of the water. **(a)** What is the density of the wood? **(b)** What happens to the block when the steel object is replaced by a second steel object with a mass less than $0.310\ \text{kg}$? What happens to the block when the steel object is replaced by yet another steel object with a mass greater than $0.310\ \text{kg}$?

10. A large balloon of mass $226\ \text{kg}$ is filled with helium gas until its volume is $325\ \text{m}^3$. Assume the density of air is $1.29\ \text{kg/m}^3$ and the density of helium is $0.179\ \text{kg/m}^3$. **(a)** Draw a force diagram for the balloon. **(b)** Calculate the buoyant force acting on the balloon. **(c)** Find the net force on the balloon and determine whether the balloon will rise or fall after it is released.

11. A spherical weather balloon is filled with hydrogen until its radius is $3.00\ \text{m}$. Its total mass including the instruments it carries is $15.0\ \text{kg}$. **(a)** Find the buoyant force acting on the balloon, assuming the density of air is $1.29\ \text{kg/m}^3$. **(b)** What is the net force acting on the balloon and its instruments after the balloon is released from the ground?

12. A man of mass $m = 70.0\ \text{kg}$ and having a density of $\rho = 1\ 050\ \text{kg/m}^3$ (while holding his breath) is completely submerged in water. **(a)** Write Newton's second law for this situation in terms of the man's mass m, the density of water ρ_w, his volume V, and g. Neglect any viscous drag of the water. **(b)** Substitute $m = \rho V$ into Newton's second law and solve for the acceleration a, canceling common factors. **(c)** Calculate the numeric value of the man's acceleration. **(d)** How long does it take the man to sink $8.00\ \text{m}$ to the bottom of the lake?

13. On October 21, 2001, Ian Ashpole of the United Kingdom achieved a record altitude of $3.35\ \text{km}\ (11\ 000\ \text{ft})$ powered by 600 toy balloons filled with helium. Each filled balloon had a radius of about $0.50\ \text{m}$ and an estimated mass of $0.30\ \text{kg}$. **(a)** Estimate the total buoyant force on the 600 balloons. **(b)** Estimate the net upward force on all 600 balloons.

14. The gravitational force exerted on a solid object is $5.00\ \text{N}$ as measured when the object is suspended from a spring scale as in **Figure P9.38a**. When the suspended object is submerged in water, the scale reads $3.50\ \text{N}$ **(Figure P9.38b).** Find the density of the object.

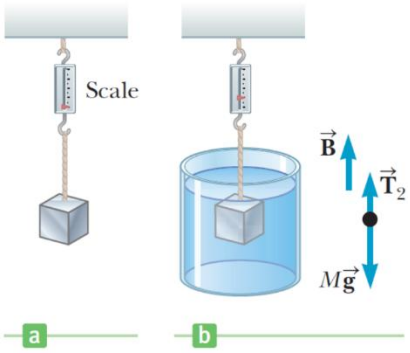


Figure P9.38

15. A cube of wood having an edge dimension of $20.0\ \text{cm}$ and a density of $650\ \text{kg/m}^3$ floats on water. **(a)** What is the distance from the horizontal top surface of the cube to the water level? **(b)** What mass of lead should be placed on the cube so that the top of the cube will be just level with the water surface?

16. A light spring of force constant $k = 160\ \text{N/m}$ rests vertically on the bottom of a large beaker of water **(Fig. P9.40a).** A 5.00-kg block of wood (density $= 650\ \text{kg/m}^3$) is connected to the spring, and the block–spring system is allowed to come to static equilibrium **(Fig. P9.40b).** What is the elongation ΔL of the spring?

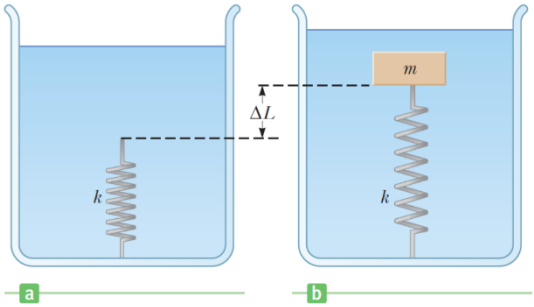


Figure P9.40

17. A sample of an unknown material appears to weigh $300\ \text{N}$ in air and $200\ \text{N}$ when immersed in alcohol of specific gravity 0.700. What are **(a)** the volume and **(b)** the density of the material?

18. An object weighing $300\ \text{N}$ in air is immersed in water after being tied to a string connected to a balance. The scale now reads $265\ \text{N}$. Immersed in oil, the object appears to weigh $275\ \text{N}$. Find **(a)** the density of the object and **(b)** the density of the oil.

19. A 1.00-kg beaker containing $2.00\ \text{kg}$ of oil (density $= 916\ \text{kg/m}^3$) rests on a scale. A 2.00-kg block of iron is suspended from a spring scale and is completely submerged in the oil **(Fig. P9.43).** Find the equilibrium readings of both scales.

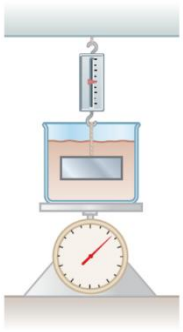


Figure P9.43

20. **(a)** Calculate the weight of a cylindrical column of water with height $h = 40.0\ \text{m}$ and radius $r = 1.00\ \text{m}$. **(See Fig. 9.6.) (b)** Calculate the force exerted by air on a disk of radius $1.00\ \text{m}$ at the water's surface. **(c)** What pressure at a depth of $40.0\ \text{m}$ supports the water column?

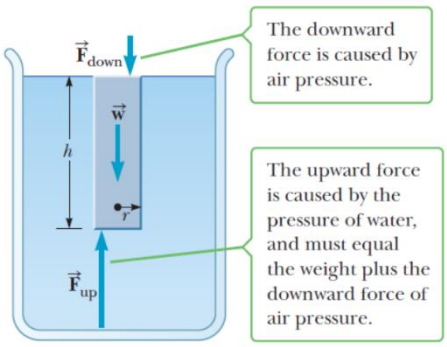


Figure 9.6 (Example 9.1)

21. A large rectangular tub is filled to a depth of $2.60\ \text{m}$ with olive oil, which has density $915\ \text{kg/m}^3$. If the tub has length $5.00\ \text{m}$ and width $3.00\ \text{m}$, calculate **(a)** the weight of the olive oil, **(b)** the force of air pressure on the surface of the oil, and **(c)** the pressure exerted upward by the bottom of the tub.

22. In a huge oil tanker, salt water has flooded an oil tank to a depth of $h_2 = 5.00\ \text{m}$. On top of the water is a layer of oil $h_1 = 8.00\ \text{m}$ deep, as in the cross-sectional view of the tank in **Figure 9.14**. The oil has a density of $0.700\ \text{g/cm}^3$. Find the pressure at the bottom of the tank. (Take $1\ 025\ \text{kg/m}^3$ as the density of salt water.)

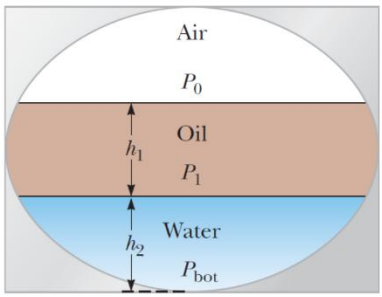


Figure 9.14 (Example 9.5)

23. Calculate the pressure on the top lid of a chest buried under $4.00\ \text{meters}$ of mud with density equal to $1.75 \times 103\ \text{kg/m}^3$ at the bottom of a 10.0-m-deep lake.

24. Estimate the net force exerted on your eardrum due to the water above when you are swimming at the bottom of a pool that is $5.0\ \text{m}$ deep.

25. An airplane takes off at sea level and climbs to a height of $425\ \text{m}$. Estimate the net outward force on a passenger's eardrum assuming the density of air is approximately constant at $1.3\ \text{kg/m}^3$ and that the inner ear pressure hasn't been equalized.

26. In a car lift used in a service station, compressed air exerts a force on a small piston of circular cross section having a radius of $r_1 = 5.00\ \text{cm}$. This pressure is transmitted by an incompressible liquid to a second piston of radius $r_2 = 15.0\ \text{cm}$. **(a)** What force must the compressed air exert on the small piston in order to lift a car

weighing $13\ 300$ N? Neglect the weights of the pistons. **(b)** What air pressure will produce a force of that magnitude? **(c)** Show that the work done by the input and output pistons is the same.

27. A bargain hunter purchases a "gold" crown at a flea market. After she gets home, she hangs it from a scale and finds its weight to be 7.84 N **(Fig. 9.24a)**. She then weighs the crown while it is immersed in water, as in **Figure 9.24b**, and now the scale reads 6.86 N. Is the crown made of pure gold?

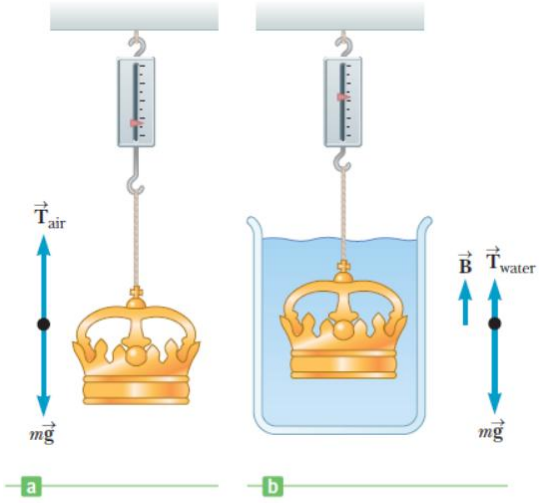


Figure 9.24 (Example 9.8) (a) When the crown is suspended in air, the scale reads $T_{air} = mg$, the crown's true weight. (b) When the crown is immersed in water, the buoyant force $\vec{B}$ reduces the scale reading by the magnitude of the buoyant force, $T_{water} = mg - B$.

28. The weight of a metal bracelet is measured to be $0.100\ 00$ N in air and $0.092\ 00$ N when immersed in water. Find its density.

29. A syringe has an area of $1.0\ \text{cm}^2$ in the barrel and then narrows down to an area of $0.10\ \text{mm}^2$ at the needle end. If a force of 5.0 N is applied to the syringe, what is the force produced at the tip of the needle?

30. A car of mass $1400\ \text{kg}$ rests on a hydraulic lift with a piston of radius $0.25\ \text{m}$ that is connected to a second piston of radius $0.030\ \text{m}$. If the car is just barely lifted off the ground, what is the force at the second piston?